ISRO

SLV Rockets:

The SLV or the Satellite Launch Vehicle was a project started in the early 1970s by ISRO (Indian Space Research Organisation) to develop the technology needed for launching the rocket. SLV was intended to reach a height of 400 kilometers and carry a payload of 40kg.This SLV rocket was a four-stage rocket with all solid-propellant motors. The first launch took place in Sriharikota on 10th August 1979.The solid motor case for first and second stage are fabricated form 15 CDV6 steel sheets and third and fourth stages from fiber reinforced plastic. This rocket has a height of 22m,diameter of 1m,mass of 17,000kg.

Flight No.	Date	Payload	Payload Mass	Orbit	Launch Outcome
E1	10th Aug 1979	Rohini Technology Payload	35kg	Low Earth	Failure
E2	18th July 1980	Rohini RS-1	35kg	Low Earth	Success
D1	31st May 1981	Rohini RS-D1	38kg	Low Earth	Partial Failure
D2	17th April 1983	Rohini RS-D2	41.5kg	Low Earth	Success

ASLV Rockets:

The ASLV, Augmented Satellite Launch Vehicle or Advanced Satellite Launch Vehicle is a five-staged solid fuel rocket developed by the Indian Space Research Organisation (ISRO) to place 150kg satellites into LEO. This project was started by India during the early 1980s to develop

technologies needed for a payload to be placed into a geostationary orbit. Its design was based on Satellite Launch Vehicle (SLV).The payloads of ASLV were Stretched Rohini Satellites. The ASLV was a five-stage vehicle. Two strap-on boosters for the first stage with the core stage igniting after booster burn out. The payload capacity of the ASLV was approximately 150kg to an orbit of 400km with 47-degree inclination. At lift-off, the ASLV generated 92,780kgf of thrust. It was a 41,000kg rocket, measuring 23.5m in length with core diameter of 1m.

Flight No.	Date	Payload	Payload Mass	Launch Outcome
D1	24th March 1987	SROSS-A	150kg	Failure
D2	13th July 1988	SROSS-B	150kg	Failure
D3	20th May 1992	SROSS-C	106kg	Partial Failure
D4	5th May 1994	SROSS-C2	113kg	Success

PSLV Rockets:

PSLV or the Polar Satellite Launch Vehicle is an expendable medium lift launch vehicle designed by ISRO (Indian Space Research Organisation).It was developed to allow India to launch its IRS (Indian Remote Sensing) satellites into Sun-synchronous orbits ,a service that was, advent of the PSLV in 1993,commercially available only from Russia. The PSLV can also launch small size satellites into Geostationary Transfer Orbit (GTO).Some notable payloads launched but PSLV include India's first lunar probe Chandrayaan-1,India's first interplanetary mission, Mars orbiter mission (Mangalyaan) and India's first space observatory ,Astrosat, World's first rocket to deploy 104 satellites-The PSLV C-37.Studies say that this vehicle is capable to deliver 600kg of payload to 550km from SHAR which began in 1978.This vehicle has 2 strap-on boosters and has a height of 44m,diameter of 2.8m which mass of average 282,000kg and has 4 stages.

Variant	Launches	Successes	Failures	Partial Failures
PSLV-G	12	10	1	1
PSLV-CA	13	13	0	0
PSLV-XL	20	19	1	0
PSLV-DL	1	1	0	0
PSLV-QL	1	1	0	0

GSLV Rockets:

GSLV or the Geosynchronous Satellite Launch Vehicle is an expendable launch system operated by ISRO (Indian Space Research Organisation).GSLV has been used in thirteen launches to date since its first launch in 2001 to the most recent on 19th December 2018 carrying the GSAT-7A military communications satellite. The GSLV project was initiated in 1990 with the objective of acquiring an Indian Launch capability for geosynchronous satellites. GSLV uses major components that are already been proven in the PSLV launchers in the form of the S125/S139 solid rocket booster and the liquid-fueled Vikas engine. The GSLV was the first rocket to use Cryogenic Engines for its flight. The GSLV rocket has a height of 49.13m, diameter of 2.8m, Mass of 414,750kg and has 3 stages.

GSLV-MK-I

Flight No.	Date	Launch site	Payload	Payload Mass	Orbit	Launch Outcome
D1	18th April 2001	1st Launch Pad of Sriharikota	GSAT-1	1540kg	GTO	Partial Failure
D2	8th May 2003	1st Launch Pad of Sriharikota	GSAT-2	1825kg	GTO	Success
F01	20th Sep 2004	1st Launch Pad of Sriharikota	GSAT-3	1950kg	GTO	Success

F02	10th Jul 2006	2nd Launch Pad of Sriharikota	INSAT-4C	2168kg	GTO	Failure

GSLV MK-II

Flight no.	Date	Launch site	Payload	Payload Mass	Orbit	Launch Outcome
D3	15th Apr 2010	2nd Launch Pad of Sriharikota	GSAT-4	2220kg	GTO	Failure
D5	5th Jan 2014	2nd Launch Pad of Sriharikota	GSAT-14	1980kg	GTO	Success
D6	27th Aug 2015	2nd Launch Pad of Sriharikota	GSAT-6	2117kg	GTO	Success
F05	8th Sep 2016	2nd Launch Pad of Sriharikota	INSAT-3DR	2211kg	GTO	Success